Towards Real-Time Precision Agriculture

A Task-Aware Multi-Modal Decision Framework

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1 Why It Matters

≈70% more food by 2050

≈70% freshwater withdrawals

30–50% nutrient loss reported

Smart-farming systems often address **one decision in isolation**, creating fragmented tools.

2 Our Contribution

ONE PIPELINE

Sensors + Images → Task-Aware ML → Farm Decisions

Irrigation **Crop** **Fertilizer** **Disease**

Unified multi-modal framework + end-to-end latency evaluation + working dashboard prototype.

3 Task-Aware Model Selection

Choose a model by balancing **error, latency and compute**.

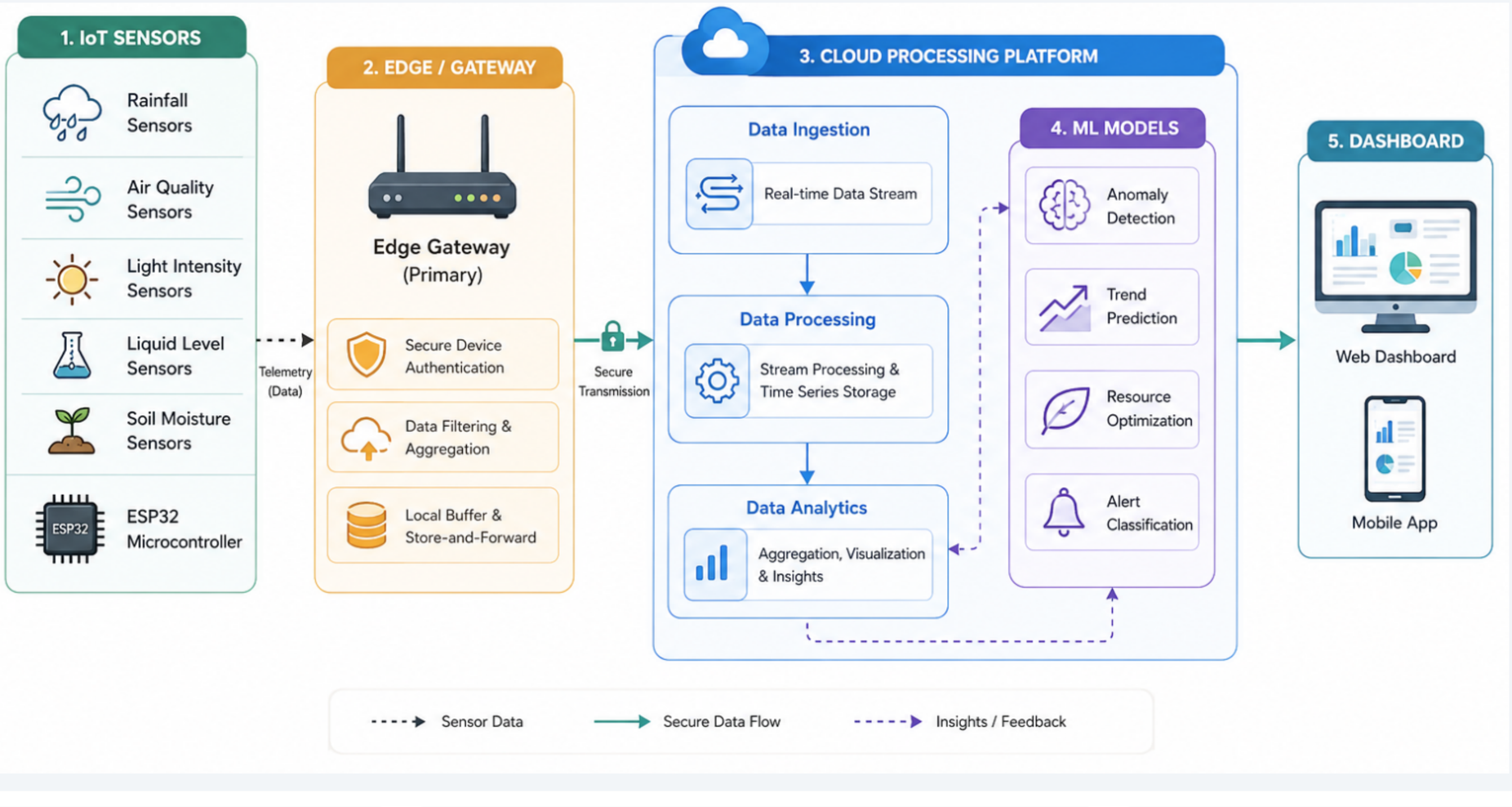
$$M^* = \arg\min_M (\alpha E + \beta L + \gamma C)$$

E Error · L Latency · C Compute

Sensor data → lightweight ML **Images** → ResNet CNN

α, β, γ are set qualitatively in this study.

4 System Architecture: from Field Sensors to Farm Decisions



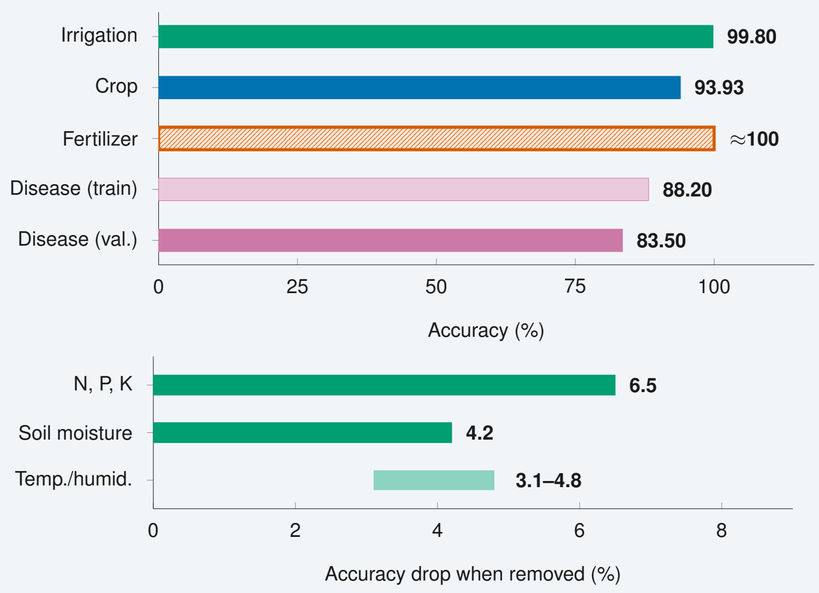
5 Datasets & Model Choices

Task	Data	Model
Irrigation	100,002 sensor samples	Decision Tree
Crop	2,200 samples	Gaussian NB
Fertilizer	**99** samples	Gaussian NB
Disease	≈87K images / 38 classes	ResNet CNN

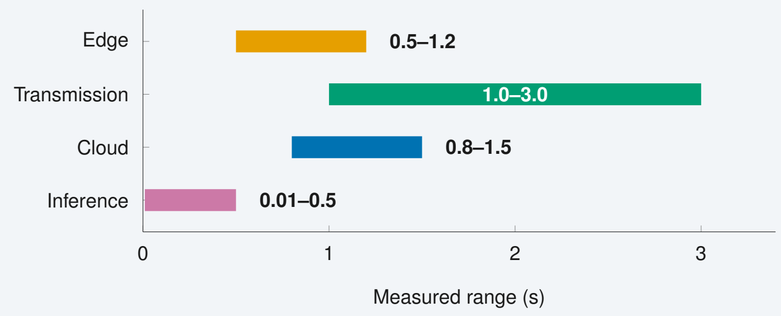
	Accuracy	Training
Random Forest	99.09%	1.74 s
Gaussian NB	93.93%	0.017 s

≈**100**× **faster training** for a 5.2-point accuracy trade-off.

6 Results: Offline Performance



7 Latency & Early Deployment

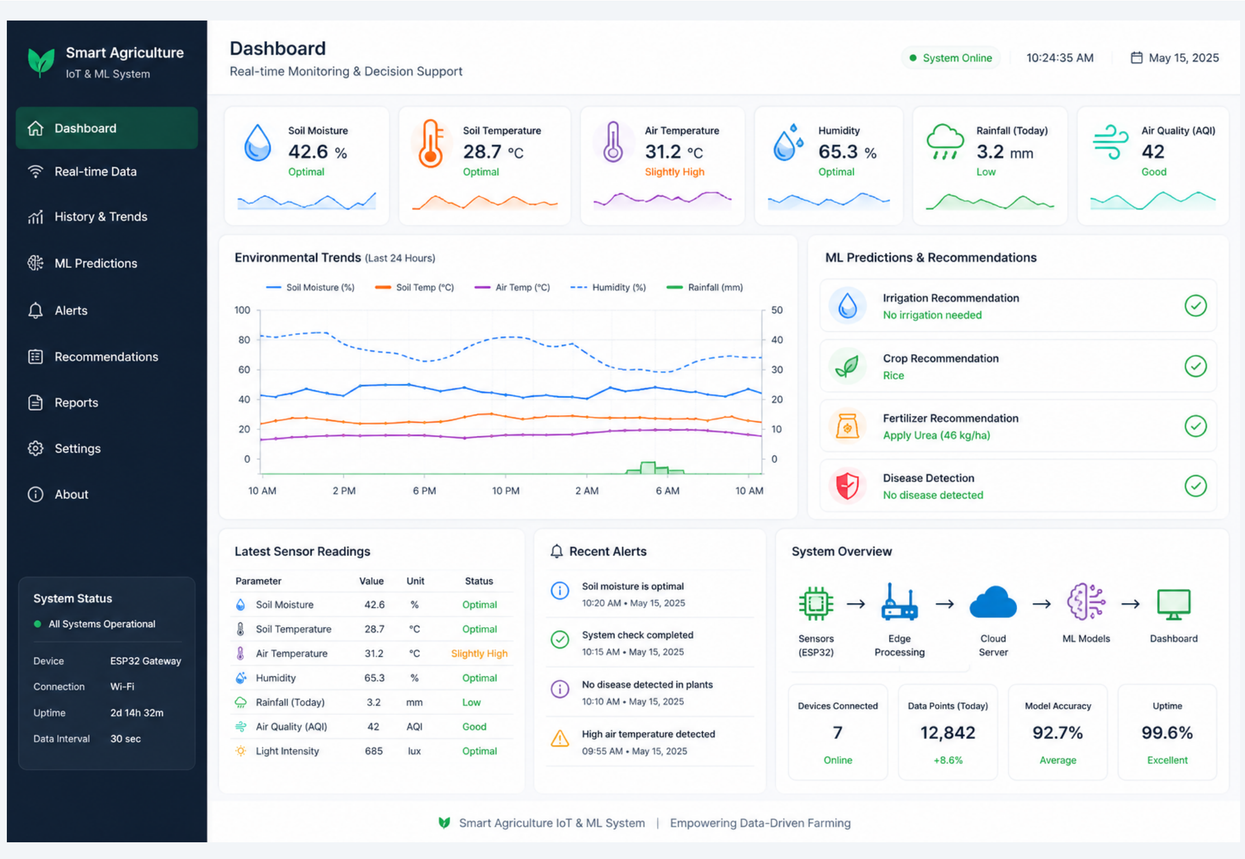


Preliminary indicators — not validated

≈22% less fertilizer **≈99.6%** uptime

Informal proof-of-concept observations. Controlled multi-site validation is planned.

8 Proof-of-Concept Prototype



9 Takeaways & Next Steps

One system. Four decisions. 3–6 s latency.
Task-matched models balance predictive performance with computational efficiency.

Current Limitations

- Fertilizer dataset: only $n = 99$
- Deployment indicators not yet controlled
- Selection weights set qualitatively
- Large-scale IoT deployment untested

Next Steps

- Controlled multi-site field trial
- Larger independent fertilizer dataset
- Optimize model-selection weights
- Per-class + broader benchmarking